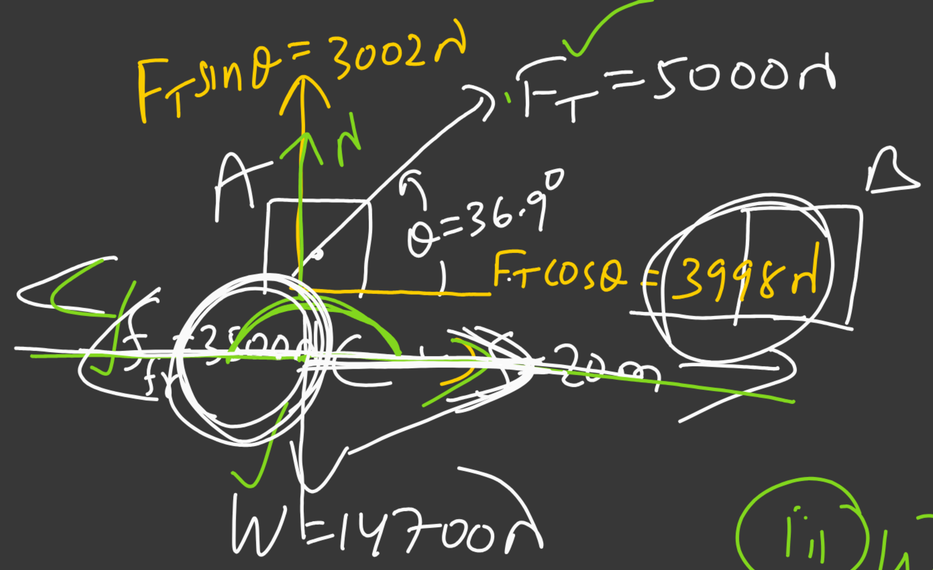
Ch-6

Ex 6.2



$$m = \frac{W}{g} = \frac{14700}{9.8} = 1500kg$$

(i) $\vec{F} \cdot \vec{S} = FS\cos\theta$

$W_{grav.} = F_{\theta=90} = 0$

$W_{Normal\ reaction} = 0$ (N, $\theta = 90$, S)

(iii) $W_{ext} = FS\cos\theta$

$= 5000 \times 20 \times \cos 36.9$

$= 80KJ$

(iv) $W_{fricl} = F_{fr} S\cos 180 = -3500 \times 20$

$= -70KJ$

$W_{Total} = 80KJ + (-70KJ) = 10KJ$

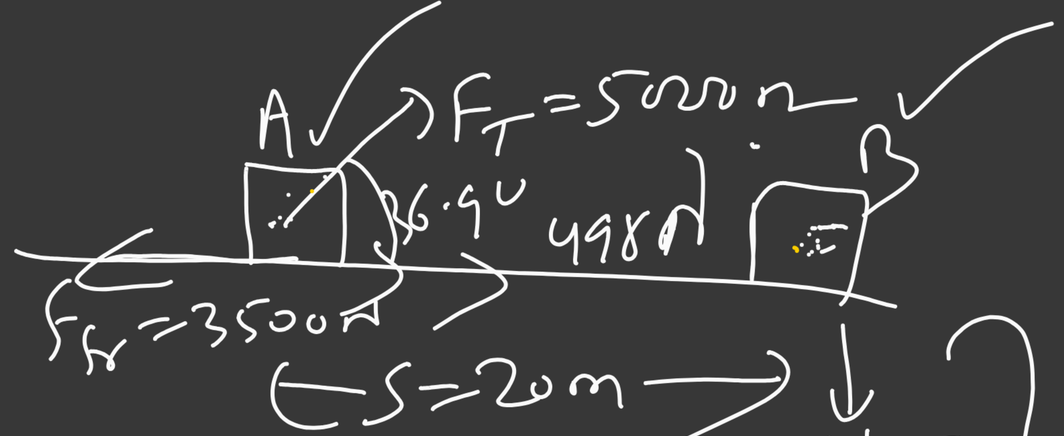
$3998 - 3500 = 498N$

$W = FS\cos\theta$

$= 498 \times 20 \times \cos 0^\circ$

$= 10KJ$

Ex 6.3



$u = 2\,m/s$

$s = 20\,m$

$a = 0.333$

$V = ?$

$V^2 - u^2 = 2as$

$V = 4.17\,m/s$

$W = \frac{1470}{9.8}$

$m = 1500\,kg$

$F_{net} = 498\,N$

$m = 1500\,kg$

$a = \frac{F}{m} = \frac{498}{1500} = 0.33\,m/s^2$

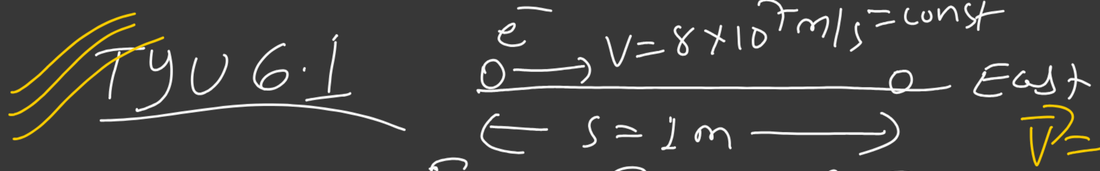
WET $W_{net} = \Delta KE$

$\Rightarrow 10{,}000\,J = \frac{1}{2}mV_f^2 - \frac{1}{2}mV_i^2$

$= \frac{1}{2} \times 1500 \left[V_B^2 - 2^2 \right]$

$\Rightarrow V_B = 4.17\,m/s$

TYU 6.1



$\vec{V}$ = const

$a = \frac{dV}{dt} = 0$

$F_{net} = ma = 0$

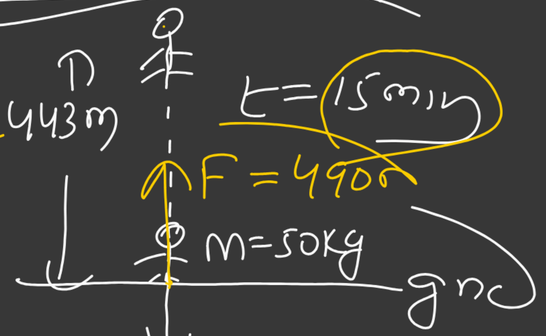
$W_{net} = F_{net} \times s$

$W_{Total\ Force} = ?$

Fgrav, Felect & Fmag

(i) +ve (ii) −ve (iii) 0 (iv) can't say

Ex 6.10



$P = \frac{dW}{dt} = \vec{F} \cdot \left(\frac{d\vec{s}}{dt}\right) = 0$

$P = \vec{F} \cdot \vec{v}$

Avg Power of man in watt, kW & hp

$W = mg = 50 \times 9.8 = 490\ N$

$W_{man} = F_{man} \times s \times \cos 0$

$W_{man} = 50 \times 9.8 \times 443$ (kg × m/s² × m)

$P = \frac{W_{Total}}{T_{Total}} = \frac{50 \times 9.8 \times 443}{15 \times 60}$ Joule/sec = watt

241 W = 0.241 kW = $\frac{241}{746}$ hp = 0.323 hp

Try 6.2 (i) 2kg body $v = 5\,m/s$ $K = \frac{1}{2} \times 2 \times 5^2 = 25J$

(ii) $m = 1\,kg$, $u = 0$ had 30J of work done on it

$K_f = ?$ $W = K_f - K_i \Rightarrow 30J = K_f - 0$ ✓

(iii) $m = 1\,kg$, $u = 4\,m/s$, 20J of Work done on it

$K_f = ?$ $W = K_f - K_i \Rightarrow 20 = K_f - \frac{1}{2} \times 1 \times 4^2$

$\Rightarrow K_f = 28J$

(iv) $m = 2\,kg$, $u = 10\,m/s$

then did 80J of work on another body

$W = K_f - K_i \Rightarrow -80 = K_f - \frac{1}{2} \times 2 \times 10^2$

$\Rightarrow K_f = -80 + 100 = 20J$

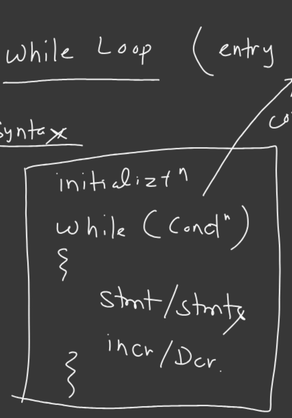
(iv) ← (i) (iii) (ii)

Looping

1. initializn
2. Condn
3. incr/Dcr.

① While Loop (entry Controlled Loop)

Syntax



Q:- Print 1 to n.

Let n = 6

o/p → 1 2 3 4 5 6



initializn

→ n

→ i = 1

Condn

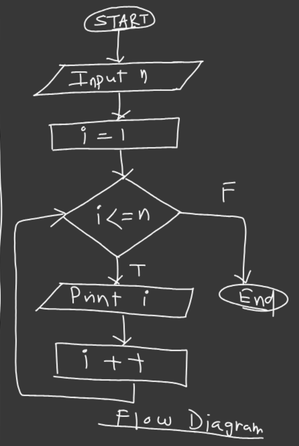
i <= n

incr/Dcr

i++

```
int n = sc.nextInt();
int i = 1;
while(i<=n)
{
  sop(i);
  i++;
}
```

java stmt



Flow Diagram

n = 4

n=4, i=1 → i<=n T Print i → i++

n=4, i=1 → 1<=4 T → 1 → 2

n=4, i=2 → 2<=4 T → 2 → 3

n=4, i=3 → 3<=4 T → 3 → 4

n=4, i=4 → 4<=4 T → 4 → 5

n=4, i=5 → 5<=4 F → End

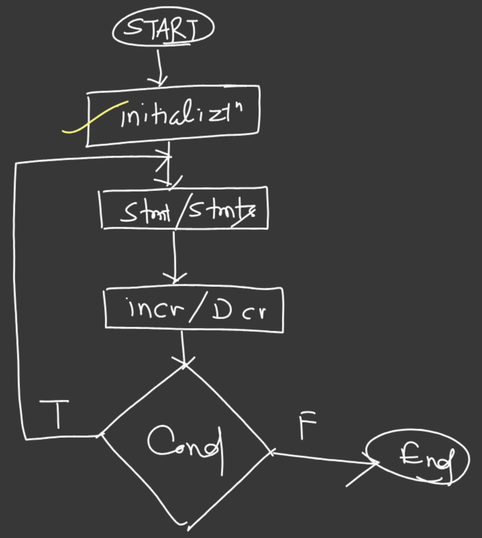
② do-while (Exit Controlled Loop) → Menu driven Program.

*Keywords :- do, while

Syntax

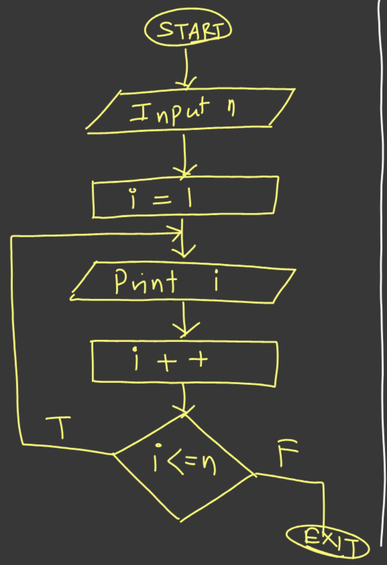
```
initialization
do
{
    stmt/stmts
    incr/Dcr
} while (Condn);
```

it Allow to execute the body of the Loop at least once even the Condn is false.



Q:- Print 1 to n.

```
int n = sc.nextInt();
int i = 1;
do
{
    s.o.pln(i);
    i++;
} while(i<=n);
```



n=4

n=4, i=1 → Print i → i++ → i<=n →

n=4, i=1 →	1	→ 2	→ 2<=4	→ T
n=4, i=2 →	2	→ 3	→ 3<=4	→ T
n=4, i=3 →	3	→ 4	→ 4<=4	→ T
n=4, i=4 →	4	→ 5	→ 5<=4	→ F

EXIT

③ for Loop (Entry Controlled Loop)

Keyword :- for

Syntax

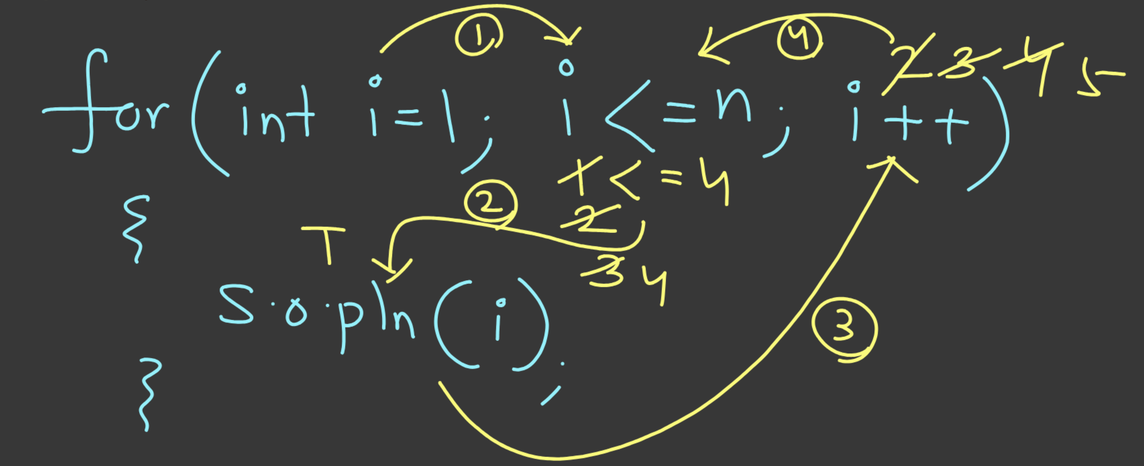
```
for ( initializtn ; Condn ; incr/Dcr )
{
    Stmt / stmts
}
```

(initializtn = ①, Condn = ②, incr/Dcr = ③)

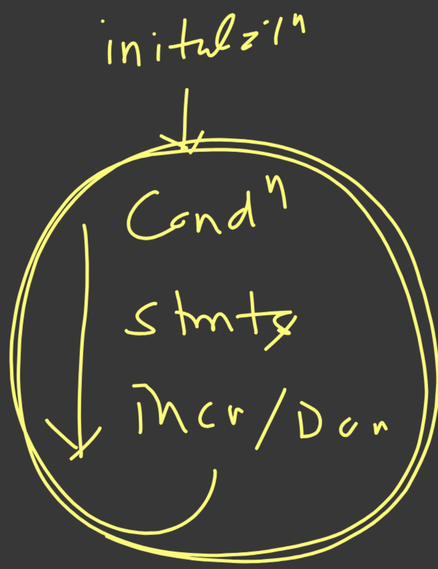
Flow Diag & exⁿ Pattern Same as while Loop.

Q :— Print 1 to n.

```
int n = sc.nextInt();
for (int i=1; i<=n; i++)
{
    S.o.pln(i);
}
```



1
2
3
4



Q:- Print All odd no. from 1 to n.

while

```
int n = sc.nextInt();
int i=1;
while(i<=n)
{
 s.o.pln(i);
 i=i+2;
}
```

do-while

```
int n = sc.nextInt();
int i=1;
do
{
  s.o.pln(i);
  i=i+2;
} while(i<=n);
```

for

```
int n=sc.nextInt();
for(int i=1; i<=n; i=i+2)
{
   s.o.pln(i);
}
```

n=10 (do-while)

n=10, i=1 → Print i → i=i+2 → i<=n

n=10, i=1 → 1 → 3=1+2 → 3<=10 → T

n=10, i=3 → 3 → 5=3+2 → 5<=10 → T

n=10, i=5 → 5 → 7=5+2 → 7<=10 → T

n=10, i=7 → 7 → 9=7+2 → 9<=10 → T

n=10, i=9 → 9 → 11=9+2 → 11<=10 → F → EXIT

DM (CSE-1002)

3.2 The growth of Functions.

$a_1 \le a_2 \dots \le a_n$

(Recall)

Searching → Linear Search : $2n+2 = \Theta(n)$

Searching → Binary Search : $2\lceil \log_2 n \rceil + 2 = \Theta(\log_2 n)$

$n = 2^2$	$n = 2^4$	$n = 2^8$
10	34	514
6	10	18

Sorting → Bubble Sort : $\frac{(n-1)n}{2} = \Theta(n^2)$

Sorting → Insertion Sort : $\frac{n(n+1)}{2} - 1 = \Theta(n^2)$

Definition (Big-O Notation, Big-Omega Notation, Big-Theta Notation)

$f, g : D \to R$ (where $(D = \mathbb{Z}) \vee (D = \mathbb{R})$)

(1) (Big-O Notation)

$f(x) = O(g(x))$

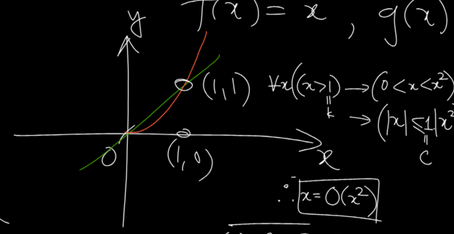
$\exists C>0\ \exists k>0\ \forall x \left((x > k) \to (|f(x)| \le C \cdot |g(x)|) \right)$

$x^2 + x \sim x^2$

$\lim_{x \to \infty} \left(\frac{x^2+x}{x^2} \right) = 1$

Eg 1 : $f, g : \mathbb{R} \to \mathbb{R}$

$f(x) = x$, $g(x) = x^2$



$\forall x \left((x > 1) \to (0 < x < x^2) \right)$

$\forall x \left((x > \underbrace{1}_{k}) \to (|x| \le \underbrace{1}_{C} |x^2|) \right)$

$\therefore x = O(x^2)$

(2) $f, g : \mathbb{R} \to \mathbb{R}$

$f(x) = g(x) = x$

$\forall x \left(|x| \le |x| \right)$

$\to \forall x \left((x > \underbrace{1}_{k}) \to (|x| \le \underbrace{1}_{C} |x|) \right)$

$\therefore x = O(x)$

(3) $x^2 + x = O(x^2)$

To prove : $\exists C>0\ \exists k>0\ \forall x \left((x>k) \to (|x^2+x| \le C|x^2|) \right)$

• $\forall x\left((x>1) \xrightarrow{k=1} (0<x<x^2)\right)$ $\quad x^2+x=O(x^2)$

$\rightarrow (x^2<x^2+x<2x^2)$ $\quad |x^2+x|\leq C\cdot|x^2|$

$\rightarrow \left(|x^2+x| \leq \underbrace{2}_{C}\cdot|x^2|\right)$

$\therefore x^2+x=O(x^2)$

(3) $x^2=O(x^2+x)$ (Hw)

(2) (Big-Omega Notation)

$f(x)=\Omega(g(x)) : g(x)=O(f(x))$

Examples (1) $x^2=\Omega(x)$

(2) $x=\Omega(x)$

(3) $x^2+x=\Omega(x^2)$

(3) (Big-Theta Notation)

$f(x)=\Theta(g(x)) : (f(x)=O(g(x)) \wedge (f(x)=\Omega(g(x))$

$a,b\in\mathbb{R}$

$a\leq b$

$a\geq b$

$\therefore a=b$

(Recall) $\log_e(\cdot) : \mathbb{R}_+ \rightarrow \mathbb{R}$

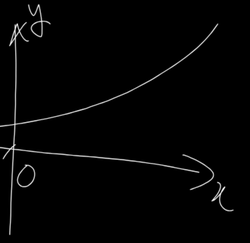


$x \rightarrow e^x$

$(\mathbb{R}) \quad (\mathbb{R}_+)$

$\rightarrow 1-1$

$\rightarrow$ Onto



• $\forall x\left((x>1) \rightarrow \left(0<\log_e(x)<x\right)\right)$

Facts: $x\log_e(x) = O(x^2)$

$\forall x\big((x>1) \rightarrow (0 < \log_e(x) < x)\big)$

$\rightarrow \big(0 < x.\log_e(x) < x^2\big)$

$\rightarrow \big(|x\log_e(x)| \leq 1|x^2|\big)$

$\therefore \boxed{x\log_e(x) = O(x^2)}$

---

(2) $3x^2 + 8x\log_e(x) = \Theta(x^2)$

$\forall x\big((x>1) \rightarrow (0 < \log_e(x) < x)\big)$

$\rightarrow \big(0 < 8x.\log_e(x) < 8x^2\big)$

$\rightarrow \big(3x^2 < 3x^2 + 8x\log_e(x) < 11x^2\big)$

$\rightarrow \big(|x^2| \leq \frac{1}{3}|3x^2 + 8x\log_e(x)|\big)$

$\big(|3x^2 + 8x\log_e(x)| \leq 11.|x^2|\big)\big)$

$\therefore x^2 = O(3x^2 + 8x\log_e(x))$

&

$3x^2 + 8x\log_e(x) = O(x^2)$

Hence Proved!

Q:- Find Factorial of a given number n.

$n! = n \times (n-1)!$

$(n-1) \times (n-2)!$

$5! = 5 \times 4!$

120

$4 \times 3!$

$3 \times 2!$

$2 \times 1!$

$1 \times 0!$

n	
0!	= 1
1!	= 1 → 1×1
2!	= 2 → 1×2
3!	= 6 → 1×2×3
4!	= 24 → 1×2×3×4
5!	= 120 → 1×2×3×4×5

$(n-2) \times (n-3)!$ n=5

1/0

→ 1 to n → Repeat.

initialization
① n
② i = 1, F = 1

Condn
i <= n

incr
i++

F = F * i
Stmt

while

```
0 int n=sc.nextInt();
  int i=1, F=1;
  while(i<=n)        // 1<=0
  {
    F=F*i;
    i++;
  }
  S.o.pln("Factorial="+F);   // 1
```

do-while

```
0 int n=sc.nextInt()
  int i=1, F=1;
  do
  { 1 = 1*1
    F=F*i;
  2 i++
  } while(i<=n);   // 2<=0
  S.o.pln("Factorial="+F);   // 1
```

for

```
0
int n=sc.nextInt();
int F=1;
for(int i=1; i<=n; i++)   // 1<=0
{
  F=F*i;
}
S.o.pln("Factorial="+F);
```

n=6, i=1, F=1 → F=F*i → i++ → i<=n

n=6, i=1, F=1 → 1 = 1*1 → 2 → 2<=6 → T

n=6, i=2, F=1 → 2 = 1*2 → 3 → 3<=6 → T

n=6, i=3, F=2 → 6 = 2*3 → 4 → 4<=6 → T

n=6, i=4, F=6 → 24 = 6*4 → 5 → 5<=6 → T

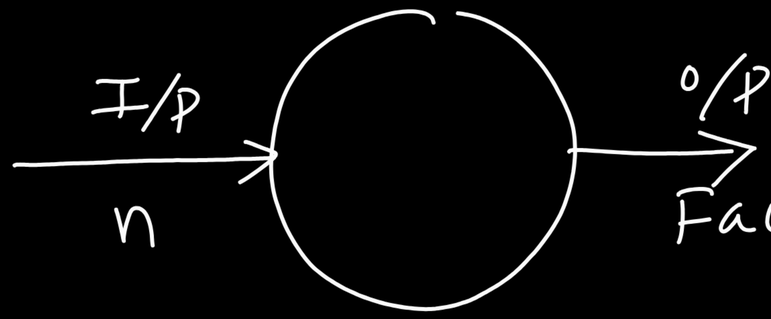
n=6, i=5, F=24 → 120 = 24*5 → 6 → 6<=6 → T

n=6, i=6, F=120 → ~~720~~ 120*6 → 7 → 7<=6 → F → 720

Checking Factorial Number or Not

Factorial Number

0! = 1
or 1! = 1
2! = 2
3! = 6
4! = 24
5! = 120
6! = 720
⋮



23 → X

24 → ✓ → 4! = 24

48 → X

120 → ✓ → 5! = 120

```
while (F < num)
{
   F = F * i
   i ++
}
```

1 = 1 * 1

2 = 1 * 2

6 = 2 * 3

24 = 6 * 4

120 = 24 * 5

⋮

F	<	num
1	<	48
1	<	48
2	<	48
6	<	48
24	<	48
120	<	48 → stop.

a/11/26

5.3 The fundamental Theorem of Calculus

Part 1 If f is continous on $[a,b]$ then the function g defined by

$$g(x) = \int_a^x f(t)\,dt \quad , a \le x \le b$$

is continous on $[a,b]$ & differentiable on (a,b) and $g'(x) = f(x)$

Part II

If f is continous on $[a,b]$, then

$$\int_a^b F(x)\,dx = F(b) - F(a)$$, where, F is any anti-derivative of f.

i.e, $F' = f$

Exer {5.3}

(13) use Part 1 of the fundamental Theorem of calculus to find the derivative of the function

$$h(x) = \int_1^{e^x} \ln t\,dt$$

let $u = e^x$

$$\frac{du}{dt} = e^x$$

$$h'(x) = \frac{dh}{dx} = \frac{dh}{du} \times \frac{du}{dx}$$

$$h'(x) = \frac{d}{dx}\left[\int_1^{e^x} \ln t\,dt\right]$$

$$= \frac{d}{dx}\left[\int_1^{u} \ln t\,dt\right]$$

$$= \frac{d}{du}\left[\int_1^{u} \ln t\,dt\right]\frac{du}{dx}$$

$$= (\ln u)e^x = (\ln e^x)\,e^x = xe^x$$ ✓

$$g(x) = \int_a^x f(t)\,dt$$

$$\frac{d}{dx}\left[\int_a^x f(t)\,dt\right]$$

$$= g'(x) = f(x)$$

Q46) what is the wrong in the eqn.?

$$\int_{-1}^{2} \frac{4}{x^3}\,dx = \frac{-2}{x^2}\Big]_{-1}^{2} = -3/2$$ → rms

AS $4/x^3$ is not continous on $[-1,2]$ the fundamental Theorem of calculus can't be applied.

Q) Find the derivative of the function.

$$g(x) = \int_{1-2x}^{1+2x} t\sin t\,dt$$

↳ rms

$$\Rightarrow \int_{1-2x}^{0} t\sin t\,dt + \int_{0}^{1+2x} t\sin t\,dt$$

$$\Rightarrow -\int_{0}^{1-2x} t\sin t\,dt + \int_{0}^{1+2x} t\sin t\,dt$$

Let $u = 1-2x \Rightarrow \frac{du}{dx} = -2$, $v = 1+2x$ $\Rightarrow \frac{dv}{dx} = 2$.

$$g'(x) = \frac{d}{dx}\left[-\int_{0}^{u} t\sin t\,dt\right] + \frac{d}{dx}\left[\int_{0}^{v} t\sin t\right]$$

$$\Rightarrow \frac{d}{du}\left[-\int_{0}^{u} t\sin t\,dt\right]\frac{du}{dx} + \frac{d}{dv}\left[\int_{0}^{v} t\sin t\right]\frac{dv}{dx}$$

$$\Rightarrow (-u\sin u)(-2) + (v\sin v)2.$$

$$\Rightarrow 2\{(1-2x)\sin(1-2x) + (1+2x)\sin(1+2x)\}$$

Q62. If $f(x) = \int_{0}^{\sin x} \frac{dt}{\sqrt{1+t^2}}$ and $g(y) = \int_{3}^{y} f(x)\,dx$, then find $g''(\pi/6)$

$$g'(y) = f(y) = \int_{0}^{\sin y} \sqrt{1+t^2}\,dt$$

$$g''(y) = f'(y) = \frac{d}{dy}\left[\int_{0}^{\sin y} \sqrt{1+t^2}\,dt\right]$$

Let $u = \sin y$, $\frac{du}{dy} = \cos y$

$$= \frac{d}{du}\left[\int_{0}^{u} \sqrt{1+t^2}\,dt\right]\frac{du}{dy}$$

$$\Rightarrow \left(\sqrt{1+u^2}\right)\cos y = \left(\sqrt{1+\sin^2 y}\right)\cos y$$

$$g''\left(\frac{\pi}{6}\right) = \left[\sqrt{1+\sin^2\left(\frac{\pi}{6}\right)}\cos\left(\frac{\pi}{6}\right)\right]$$

$$\Rightarrow \left[\sqrt{1+\left(\frac{1}{2}\right)^2}\right]\frac{\sqrt{3}}{2} = \frac{\sqrt{15}}{4}.$$

HW $\Rightarrow$ 5.3 Q 7, 12, 14, 28, 38, 55, 63.

5.4 Indefinite Integrals & the Net change theorem

$$\int F(x)\,dx = F(x) \Rightarrow F(x) = F'(x)$$

Application.

The Net change Theorem

The Integral of a rate of change is the net change $\int_{a}^{b} f'(x)\,dx = F(b) - F(a)$

Exercise 5.4

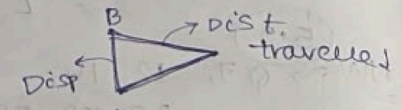
Q54) Find the general indefinite integral

$$\int (\theta - \operatorname{cosec}\theta \cdot \cot\theta)\, d\theta =$$

$$\Rightarrow \frac{\theta^2}{2} + \operatorname{cosec}\theta + C.$$

Q59) The velocity function in m/sec is given for a particle moving along a line a) Find the displacement (b) The distance travelled by the particle during the given time interval.

$$V(t) = 3t - 5, \quad 0 \le t \le 3$$



$a = 0$, $b = 3$, $c = 5/3$

$$V(t) = s'(t)$$

$$s(t) = \int_0^3 (3t - 5)\, dt$$

$$\left[\frac{3t^2}{2}\right]_0^3 - \left[5t\right]_0^3$$

$$\frac{27}{2} - \frac{15}{1} = \frac{27 - 30}{2} = \frac{-3}{2}$$

$$3 \times \frac{9}{2} - 0 - 5 \times 3$$

(b) Dist travelled.

$$\int_0^3 |3t - 5|\, dt.$$

Ans)

$$\int_0^{5/3} (5 - 3t)\, dt + \int_{5/3}^3 (3t - 5)\, dt$$

$$\Rightarrow \left[5t\right]_0^{5/3} - \left[\frac{3t^2}{2}\right]_0^{5/3} + \left[\frac{3t^2}{2}\right]_{5/3}^3 - \left[5t\right]_{5/3}^3$$

$$5 \times \frac{5}{3} - \frac{3 \times 25}{2 \times 9} + \frac{3 \times 9}{2} - \frac{3 \times 25}{2 \times 9} - 5 \times 3 - 5 \times \frac{5}{3}.$$

$$\frac{25}{3} - \frac{25}{2} + \frac{27}{2} - \frac{25}{2} - 15 - \frac{25}{3}.$$

$$= \frac{41}{6}.$$

Q64) Water flows from the bottom of a storage tank at a rate of $r(t) = (200 - 4t)$ lt/min, $0 \le t \le 50$ find the amt of water that can flow from the tank during the first 10 mints

$$\int_0^{10} (200 - 4t)\,dt$$

Ans $\left[200t - \frac{4t^2}{2}\right]_0^{10}$

$200 \times 10 - \frac{4 \times 100}{2}$

$2000 - 200$

1800 ltr

HW (18, 32, 60, 61, 69.)